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High Permeability Streaks Characterization Based on AI and Full Frequency Seismic Inversion in Middle East Carbonate Reservoirs

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Abstract

Accurately predicting high-permeability streaks (HPS) in carbonate reservoirs is crucial for enhancing efficiency and recovery rates in oil and gas field development. Addressing the challenge of predicting HPS in carbonate reservoirs, especially critical for water injection in the Middle East. To improve the accuracy of HPS prediction, a new integrated HPS characterization method has been proposed. This method combines thin section with core porosity and permeability tests to identify HPS by integrated core facies identification. Subsequently, advanced AI assisted petrophysical reservoir type (AI-PRT) analysis techniques are employed to predict HPS in wells and confirmed by core facies analysis. To surpass traditional method limitations, high-resolution full frequency seismic inversion (FFI) is used to delineate 3D HPS prediction based on 4 kinds of different frequency seismic inversion. Finally, 3D permeability modelling is established by machine learning method based on the AI-PRT and high-resolution FFI. This HPS characterization method has been successfully applied in the W oilfield in the Middle East. Advanced AI-assisted petrophysical reservoir type (AI-PRT) analysis techniques are helpful for the accurate identification of HPS in well logging. The high-resolution full frequency seismic inversion (FFI) technology is conducive to establishing 3D HPS model with higher resolution. Based on the 3D HPS model, four typical HPS in W oilfield were characterized by this method: ① HPS controlled by shoals, mainly distributed in high parts of the paleo-structure; ② HPS controlled by the progradation shoals, mainly develops in the slope area, these HPS are the main potential area of lithological reservoir; ③ HPS controlled by tidal channels mainly develop in formation with tidal flats deposited. This kind of HPS easily caused water injection breakthrough; ④ HPS formed by exposed and erosion, mainly develops at the top of the high-level system and the high parts of the paleo-structure. The accurate identification of four types of HPS in carbonate reservoirs provides a scientific basis for the optimal design of well locations and efficient water injection development plan. This method realizes the identification of HPS in wells without core by advanced AI-PRT analysis techniques. 3D HPS distributions were predicted by high-resolution full-frequency inversion. The accuracy of 3D HPS model has been significantly improved based on the advanced AI-PRT and high-resolution FFI. This HPS characterization method is applicable not only to carbonate reservoirs but also to clastic reservoirs.

Introduction

To accurately predict high-permeability streaks in reservoirs is crucial for enhancing the efficiency and recovery rates of oil and gas fields. These streaks serve as optimal channels for fluid migration, directly influencing the effectiveness of injection-production systems and overall development outcomes. By precisely identifying these streaks, optimal well network deployment can be achieved, injection-production ratios adjusted, and development strategies such as water or gas flooding formulated. This approach minimizes ineffective injections, thereby maximizing development benefits.

Additionally, predicting high-permeability streaks helps mitigate early water or gas breakthrough risks, delaying natural field decline and effectively extending the development lifecycle. Particularly in highly heterogeneous carbonate reservoirs, high-permeability streaks often represent critical "sweet spots," and their accurate identification plays a decisive role in efficient development (Xiaomin, 2013).

Various methods are currently employed to predict these streaks, including core analysis, logging interpretation, seismic prediction, well testing, and dynamic data computations. Each method offers distinct advantages and limitations, tailored to different geological conditions.

Core analysis provides direct permeability measurements, enhanced in recent years by high-precision scanning technologies like CT scans and Digital Rock Physics (DRP). Despite its high accuracy in direct measurements, core analysis suffers from poor spatial representation of reservoir-wide permeability distribution and high costs associated with lengthy experimental cycles (Babak, 2016; Jianguo, 2021; Pirrone, 2024).

Logging data (such as resistivity, density, and sonic travel time) are used for permeability estimation, employing methods like empirical formulas, crossplot techniques, Formation Micro-Imaging (FMI) (Kun Meng, 2024), and Nuclear Magnetic Resonance (NMR) logging (Di Jianwei, 2016). In recent years, machine learning and deep learning have been instrumental in establishing nonlinear mappings between logging data and permeability, with NMR logging directly reflecting pore structures and achieving high precision (Liuqing Yang, 2023; Parra, 2022). However, logging interpretation heavily relies on empirical formulas, limiting regional adaptability, and indirect estimation methods are sensitive to reservoir complexities and require accurate calibration with core data.

Seismic data (including impedance and elastic parameters) predict reservoir permeability through correlation analyses, employing methods such as Amplitude Variation with Offset (AVO) analysis, seismic attribute fusion, elastic wave inversion, and machine learning-integrated multi-attribute analysis (Qian Ding, 2023; DING Qian, 2023; Miele, 2023; Lideng, 2019; Mahaman Salifou, 2022; Guohua Wei, 2021, Al-Anazi, 2010). Despite its broad spatial coverage and ability to characterize large-scale permeability distributions, seismic methods suffer from low vertical resolution and limited accuracy in depicting thin-layer permeability variations. Moreover, prediction outcomes are dependent on empirical models, introducing significant uncertainties.

Production dynamic data analysis utilizes material balance equations, flow simulations, and historical fitting to infer permeability distributions (Shangbin, 2005). Recent advancements in AI-based production data analysis have shown enhanced capability in capturing nonlinear characteristics, reflecting dynamic permeability changes over time and facilitating integration with reservoir numerical simulations (Raheem, 2024). However, results are sensitive to data quality and parameter assumptions, and the method struggles to directly characterize reservoir heterogeneity.

No single method can accurately predict the distribution of high-permeability reservoir streaks. To enhance prediction accuracy, researchers worldwide are integrating multidisciplinary approaches, including core analysis, logging interpretation, seismic attribute analysis, geological modeling, and numerical simulations. This multidisciplinary fusion approach aims to achieve more precise spatial distribution and physical property characterization of high-permeability streaks but faces challenges in integrating diverse data sources and matching cross-scale information (Pirrone, 2024; Kun Meng, 2024). Drawing from cutting-

edge international research, we have explored the geological mechanisms of high-permeability streaks and innovatively proposed a multisource data fusion method for predicting these streaks. This method leverages AI technologies to organically integrate and collaboratively invert multidimensional information from core analysis, logging interpretation results, seismic inversion attributes, well testing outcomes, and dynamic production data. Thus, it achieves accurate three-dimensional spatial predictions of high-permeability streaks in reservoirs.

Method

Highpermeability streaks (HPS) play a crucial role in hydrocarbon reservoir development, influencing production efficiency and recovery rates. Traditional methods for identifying HPS often face limitations in accuracy and spatial resolution. This paper introduces a novel integrated approach that combines advanced geological and geophysical techniques to enhance the characterization and prediction of HPS in reservoirs.

Core facies analysis based on the integrated analysis

Integrated core facies identification: thin section examination, core facies analysis, well log, and seismic data are combined to identify geological facies and their spatial distribution within the reservoir. The advantage of this method is more applicable in the subsequent well logging facies analysis and seismic facies analysis (Figure 1).

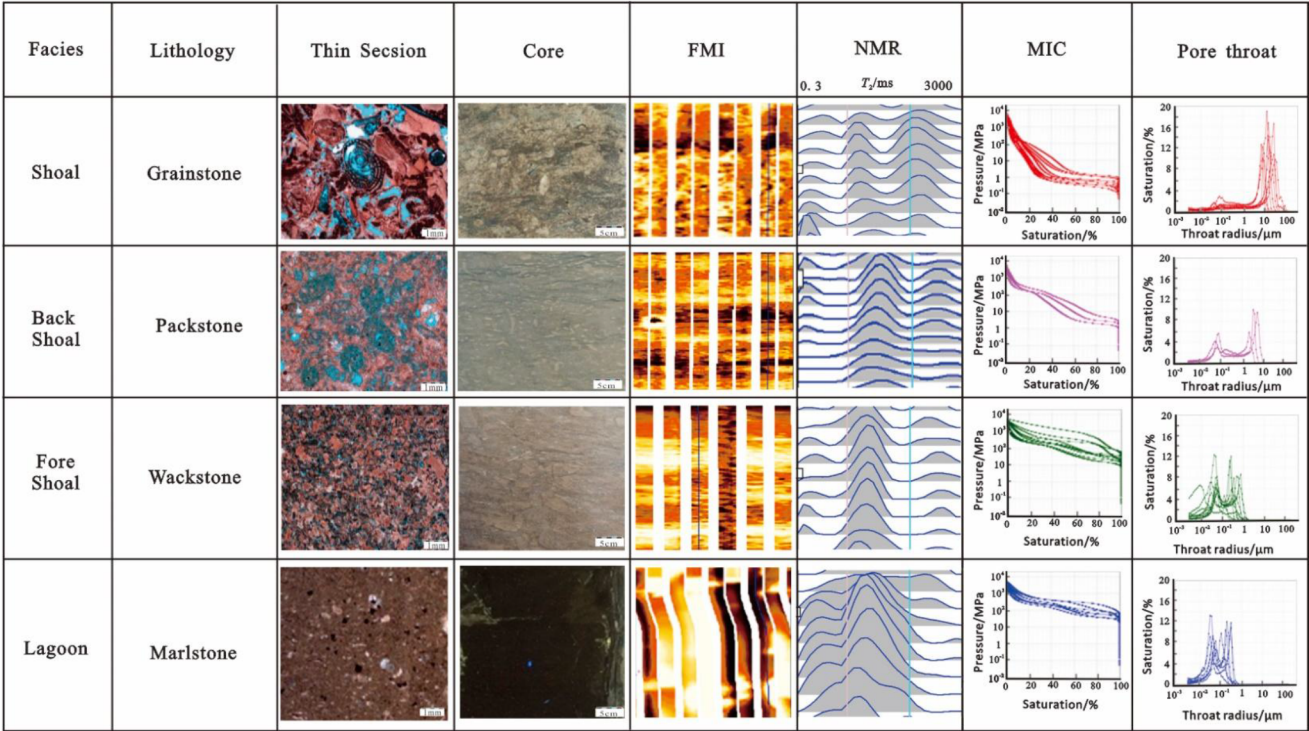


Figure 1—Integrated core facies identification

AI-Assisted Petrophysical Reservoir Type (AI-PRT) Analysis

To scale up the core observations and extend characterization to uncored wells, we implement an AI-assisted approach for petrophysical reservoir typing (Figure 2). The following steps are emphasized:

- 1. **Data Preparation:** Standard well logs (such as GR, RHOB, NPHI, DT, and resistivity) from cored and uncored wells are normalized and quality-controlled. AI well log condition was applied to make

sure the well logs have high quality. Cores serve as labeled examples, while logs provide the features for classification.

2. **Algorithm Selection and Training:** Supervised machine learning algorithms, including support vector machines (SVM), random forests, and deep learning models, are trained using the labeled core-reservoir types. Deep learning models are usually regarded as the models with the highest accuracy in logging facies analysis. Feature importance analysis is conducted to determine which log attributes most strongly influence lithology classification.
3. **Reservoir Type Classification:** The AI-PRT analysis classifies each depth interval into reservoir types with distinct petrophysical signatures. Types associated with high-permeability (e.g., grain-dominated or vuggy facies) are flagged as probable HPS zones.
4. **Cross-Well Prediction:** Once trained, the AI model is deployed across the entire well set. It enables consistent reservoir typing in wells lacking core data, thus extending HPS predictions laterally and vertically.

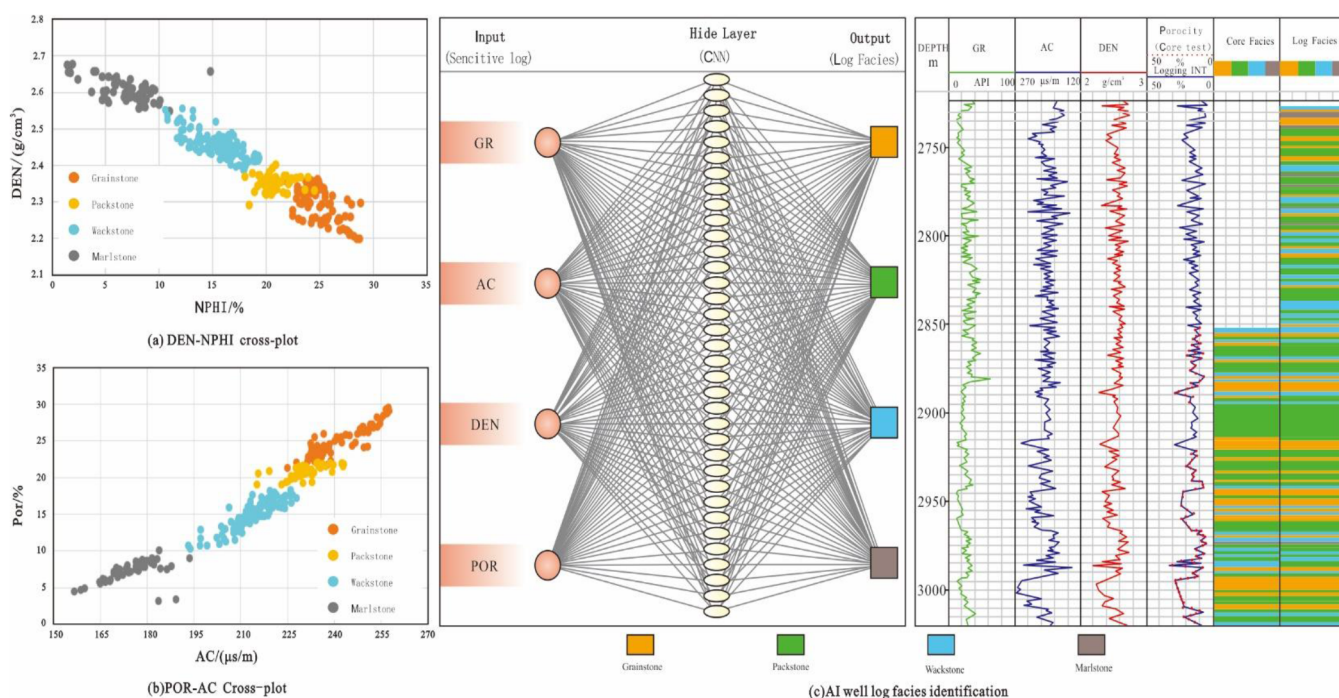


Figure 2—Sensitive well log analysis and AI well log facies identification: (a) DEN-NPHI cross-plot; (b) POR-AC Cross-plot; (c) AI well log facies identification and core facies verification

High-Resolution Full-Frequency Seismic Inversion (FFI)

While AI-PRT provides well-based HPS identification, extending this characterization to the inter-well space requires seismic information. Full-frequency inversion (FFI) offers a breakthrough by extracting high-resolution rock property volumes from seismic data (Fig. 3).

1. **Seismic Data Conditioning:** The inversion begins with pre-stack time-migrated seismic gathers, which are conditioned through noise attenuation, amplitude balancing, and phase correction.
2. **Multi-Frequency Inversion Strategy:** Instead of relying on a single-frequency band, FFI divides the seismic inversion into four frequency bands: low, mid-low, mid-high, and high frequencies. Each band captures different geological scales—from large-scale bedding to fine-scale streaks.
3. **Inversion Execution:** Each band is inverted separately using model-based or statistical methods (such as sparse spike or recursive inversion). The inverted volumes (P-impedance, V_p/V_s ratio, density) are then recombined to form a full-frequency high-resolution seismic cube.

4. **Calibration with AI-PRT:** HPS zones identified through AI-PRT are mapped onto the seismic domain. The corresponding seismic response characteristics—such as high amplitude, low impedance, or anomalous frequency content—are extracted as seismic indicators of HPS.
5. **Seismic Attribute Extraction:** Additional attributes such as coherence, curvature, spectral decomposition, and instantaneous frequency are computed to enhance the detection of subtle lateral continuity in HPS zones.

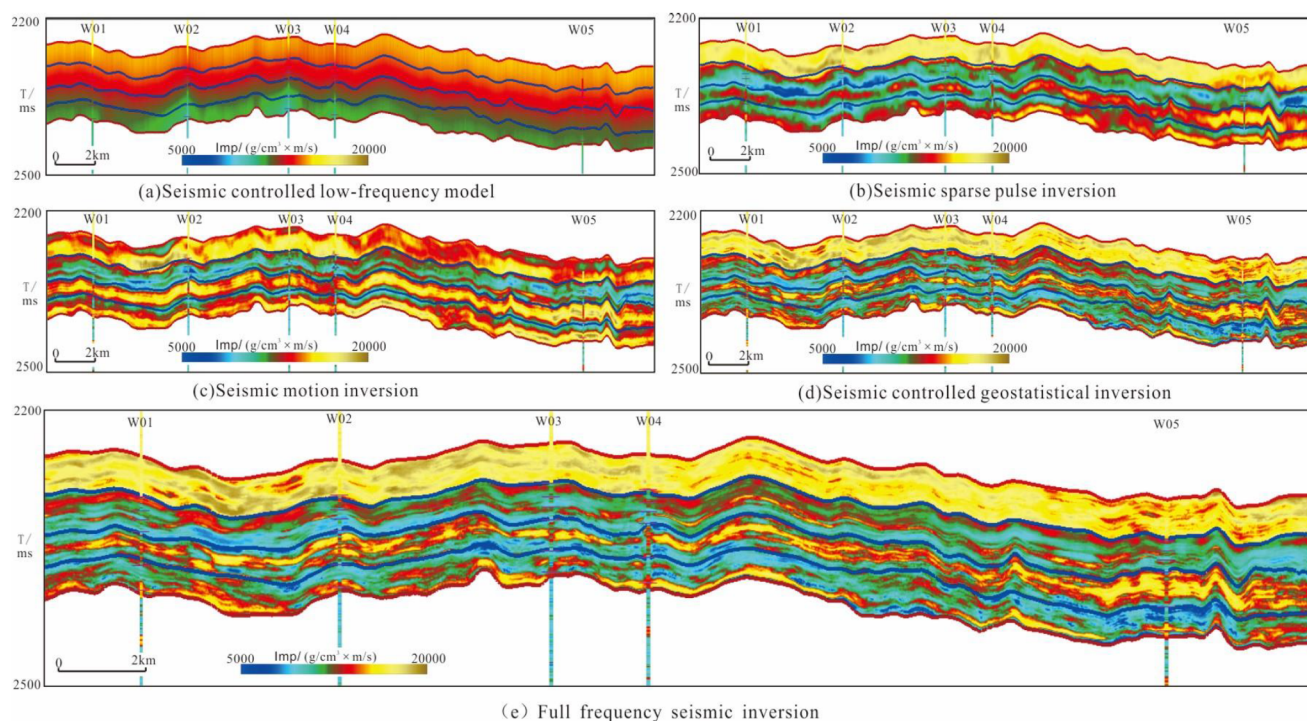


Figure 3—Comparison of four kinds of different frequency seismic inversion and full frequency seismic inversion

This multi-frequency approach significantly improves resolution, enabling reliable detection of thin, laterally extensive permeability streaks.

3D Permeability Modeling using Machine Learning

The final step in the workflow is to integrate AI-PRT predictions and FFI seismic inversion using machine learning techniques to build a robust 3D permeability model (Figure 4).

1. **Feature Engineering:** Key inputs include seismic attributes (from FFI), well log-based petrophysical types (from AI-PRT), depth, structural position, and distance to major faults. These features are collated into a spatially registered training dataset.
2. **Model Selection:** Ensemble methods like gradient boosting machines (GBM) and neural networks are tested for predictive performance. The choice of algorithm is based on cross-validation and prediction accuracy against known core permeability values.
3. **Model Training and Validation:** The ML model is trained on wells with both AI-PRT and FFI data. Predicted permeability is validated against blind wells, and error metrics (MAE, RMSE) are calculated.
4. **3D Permeability Volume Generation:** Once validated, the trained model is used to populate a 3D geo-cellular grid with permeability values across the reservoir. This grid highlights the spatial distribution of high-permeability streaks and their connectivity patterns.

5. **Visualization and Interpretation:** The 3D permeability model is visualized in reservoir modeling software. Connectivity analysis and flow simulation modules are used to assess the impact of HPS on fluid flow, sweep efficiency, and breakthrough behavior.

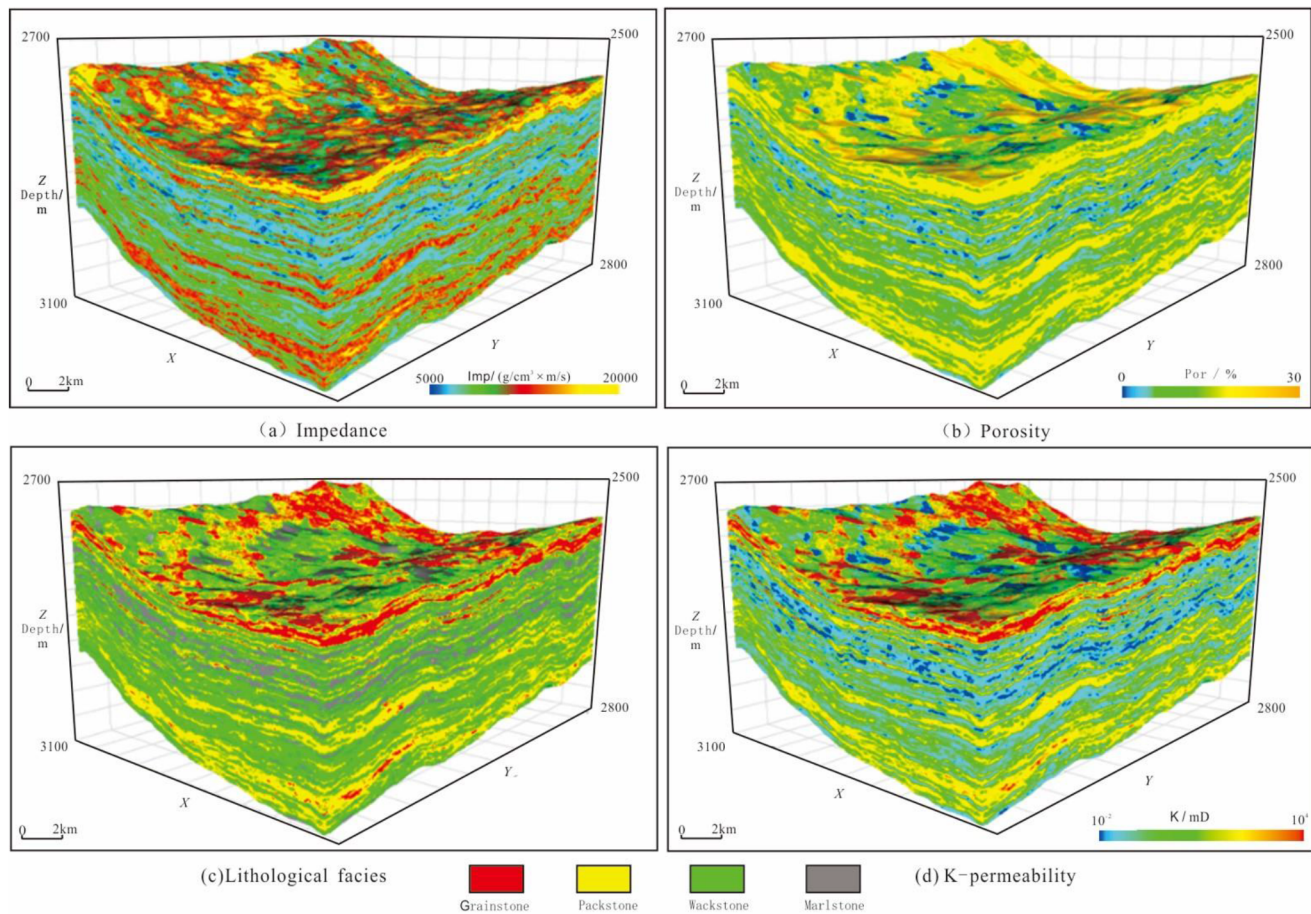


Figure 4—Reservoir properties modeling based on high quality full frequency seismic inversion

This modeling step provides not just static prediction but a dynamic basis for reservoir simulation and optimization.

Field Application: Case Study in the W Oilfield, Middle East

The new integrated High Permeability Streaks (HPS) characterization method has been successfully applied in the W oilfield, one of the most representative carbonate reservoirs in the Middle East. The field is characterized by complex depositional environments, significant diagenetic overprint, and strong heterogeneity—conditions under which traditional reservoir evaluation techniques often struggle to provide reliable HPS prediction and mapping.

Background of the W Oilfield

The W oilfield is a giant carbonate field composed of a thick succession of shallow marine sediments deposited in a ramp setting. Reservoir quality is primarily controlled by facies, diagenesis, and fracture development. High-permeability streaks, although often limited in vertical thickness (<5m), exert a disproportionate impact on production performance due to their strong influence on fluid flow pathways and sweep efficiency. Historically, the field faced challenges in accurately mapping HPS due to the limited resolution of seismic inversion products and the scarcity of core data in many wells. This made it difficult

to delineate flow units, understand water breakthrough mechanisms, and optimize well placement. The integrated workflow proposed in this study was applied to address these challenges and to improve both static and dynamic reservoir models.

AI-PRT Analysis for Log-Based HPS Identification

One of the key breakthroughs of this approach is the use of advanced AI-assisted petrophysical analysis to classify reservoir types directly from well logs, thereby enabling the prediction of HPS in wells without core data.

In the W oilfield application, a set of 70 wells was selected for AI-PRT model training and validation. Among them, 7 wells had core and thin section data, which were used to label intervals according to core facies and permeability behavior. These intervals were assigned to distinct reservoir types using a combination of porosity, permeability, pore structure, and facies descriptions.

The AI-PRT model employed a combination of ensemble deep learning and deep neural networks to process conventional logs (GR, RHOB, NPHI, DT, resistivity) and predict reservoir types corresponding to known HPS intervals. The model achieved a classification accuracy of over 91% in blind test wells, confirming its robustness and reliability.

By extending the AI-PRT predictions to uncored wells, the analysis allowed for continuous HPS prediction across the entire well set. Importantly, the model not only flagged high-quality reservoir intervals but also captured subtle trends in permeability enhancement associated with diagenetic overprinting and facies transitions.

High-Resolution FFI for 3D HPS Delineation

The second major component of the workflow involved the application of full-frequency seismic inversion (FFI) to generate high-resolution seismic property volumes capable of resolving subtle HPS geometry and connectivity.

In the W oilfield, separate inversions were conducted for each band, generating impedance and V_p/V_s volumes with different vertical resolution characteristics. These individual inversion results were then integrated using a weighted frequency fusion algorithm, producing a full-frequency seismic impedance cube with enhanced resolution and signal continuity.

The FFI-derived volumes were calibrated against the AI-PRT results in wells, where HPS intervals identified by logs were correlated with specific seismic signatures—typically low acoustic impedance, strong negative reflection amplitude, or high spectral energy in high-frequency bands.

This calibration enabled the development of seismic indicators that reliably reflect the presence of HPS in the inter-well space. Compared to traditional inversion methods, the FFI approach provided at least a 27% improvement in vertical resolution, significantly enhancing the interpretability of thin permeability streaks.

Classification of Four Types of HPS in the W Oilfield

Based on the integration of AI-PRT results and the high-resolution FFI cube, a 3D HPS prediction model was constructed. This model successfully identified four major types of HPS in the W oilfield, each with distinct genesis, distribution, and development implications (Fig.5).

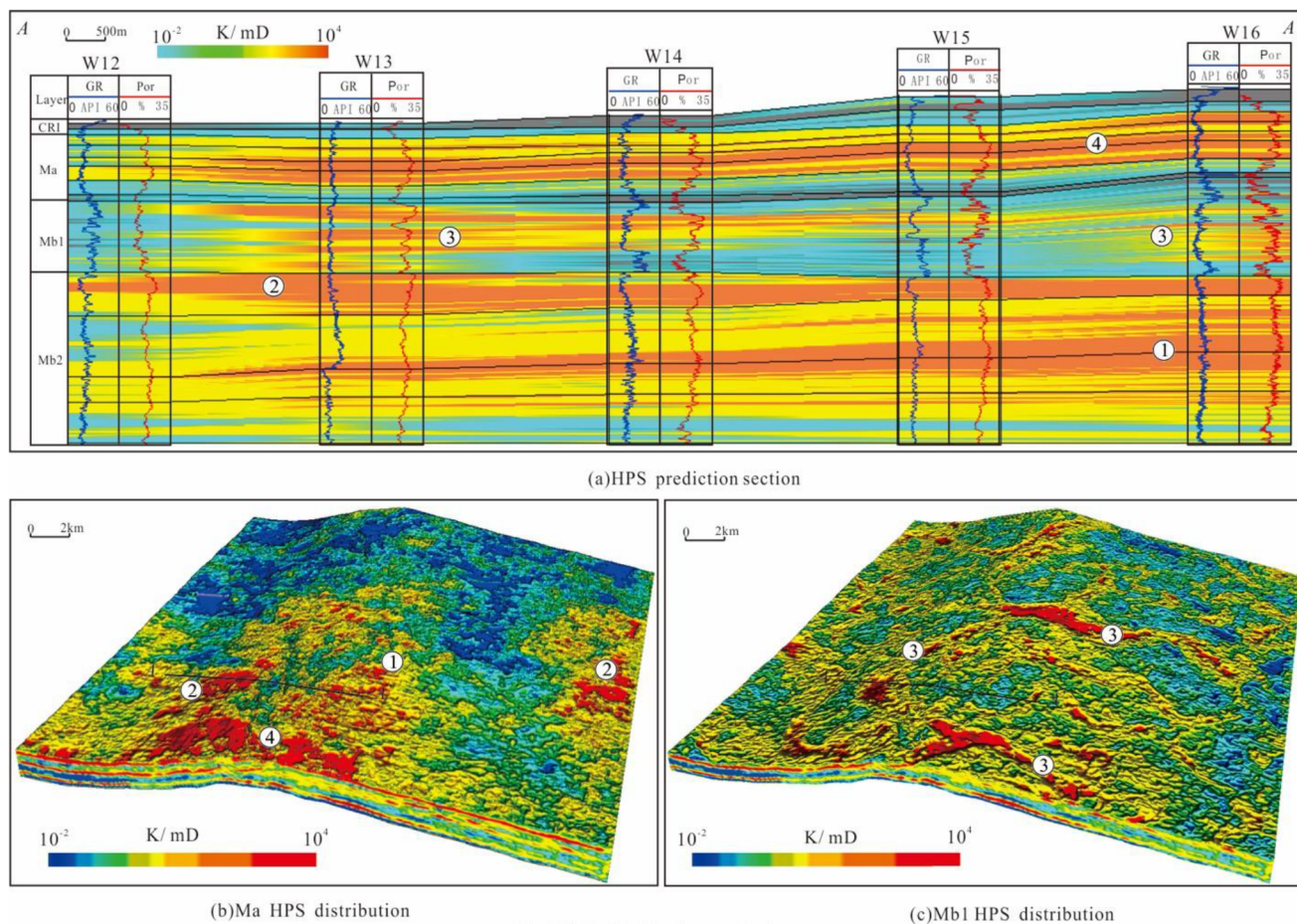


Figure 5—HPS distribution prediction. (① - HPS Controlled by Shoal Deposits, ② - HPS Formed by Progradation Shoals, ③ - HPS Associated with Tidal Channels, ④ - HPS Related to Subaerial Exposure and Erosion)

HPS Controlled by Shoal Deposits (Fig. 5-①). These streaks are primarily distributed along the structural crests of the reservoir, where grain shoals were originally deposited in high-energy shallow marine environments. They are characterized by coarse grain size, interparticle porosity, and excellent lateral continuity. The permeability of this type of HPS often exceeds 2000 mD. Due to their location in paleo-highs, these streaks often coincide with oil-bearing zones and are key targets for production wells.

HPS Controlled by Shoal Deposits (Fig. 5-②). These HPS are found in the slope zones adjacent to the main platform. They result from the lateral accretion and progradation of grain shoals during highstand periods. These HPS tend to be thinner and more segmented but still exhibit permeability in the range of 800–1500 mD. They often form the backbone of lithological traps and represent underdeveloped oil zones with significant remaining potential. Identifying these zones is critical for targeting lateral and step-out infill drilling. These kinds of HPS are very important potential lithology reservoirs in slope area.

HPS Associated with Tidal Channels (Fig. 5-③). Developed within intertidal to supratidal settings, these HPS are associated with erosive tidal channels and are typically localized within tidal flat facies. Their permeability can be extremely high (>2500 mD), but their distribution is highly irregular. These streaks are notorious for causing early water breakthrough in injection wells due to their high connectivity and alignment with injection-to-production pathways. Recognizing and managing these HPS are essential for waterflood control and optimization. These kinds of HPS are very complex in distribution.

HPS Associated with Tidal Channels (Fig. 5-④). These streaks occur at the tops of high-frequency sequences and structural highs, where subaerial exposure and erosion have enhanced secondary porosity

through dissolution processes. They are usually developed in the uppermost part of the reservoir and can be thin but pervasive. These HPS often act as vertical fluid conduits and are important for understanding cross-flow behavior between stratigraphic intervals.

This classification of HPS types provided the basis for targeted development strategies—ranging from production well optimization in shoal-controlled HPS to water control measures in channel-associated streaks.

Summary

In summary, the integrated HPS characterization method has 4 advantages: 1) Enabled accurate identification of HPS in both cored and uncored wells using AI-PRT. 2) Provided high-resolution 3D imaging of HPS distribution using full-frequency seismic inversion. 3) Delivered a 3D permeability model with improved accuracy and direct impact on field development outcomes. 4) Characterized four geologically distinct types of HPS, supporting targeted reservoir management strategies.

This case study confirms the method's value in enhancing both the static characterization and dynamic understanding of high-permeability features, making it a powerful tool for next-generation reservoir development workflows.

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